

Lab: Secure AWS Account Setup + Free Tier EC2 Deployment

Objective: Set up a production-ready secure AWS account, understand infrastructure choices, apply Shared Responsibility, stay within Free Tier economics, while demonstrating core cloud computing principles.

Skills Covered:

- IAM & Account Security
- Billing & Cost Control
- Global Infrastructure (Region + AZ awareness)
- Shared Responsibility Model
- Cloud Economics (Free Tier + alerts)
- Console navigation & basic resource deployment

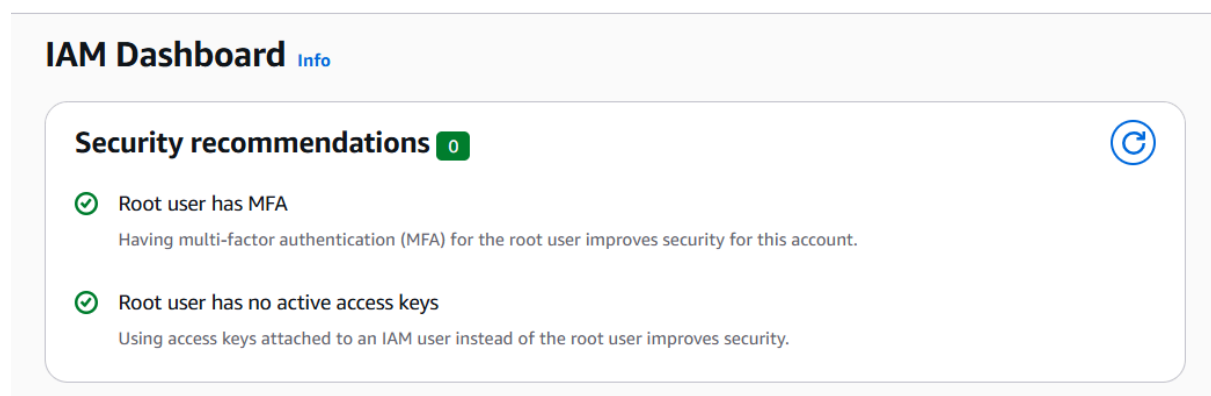
Pre-Lab Checklist

1. Log in as **Root User** (email + password)
2. Switch to **us-east-1** (N. Virginia) — critical for billing

Phase 1: Account Security & Billing Setup

Step 1: Enable MFA on Root Account (Non-negotiable)

- Go to **IAM** → Your account name (top right) → **Security credentials**
- Under **Multi-factor authentication (MFA)** → Add MFA
- Use a virtual MFA app (Google Authenticator / Authy)



Step 2: Create an IAM Admin User

- IAM → Users → **Create user**
 - User name: YourName-Admin
 - Check **Provide console access**

- Console password: Custom (temporary)
- Next → Attach policies: **AdministratorAccess** (for learning)

The screenshot shows the AWS IAM console for the user 'SimeonLab-Admin'. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, Access reports, and CloudShell. The main content area displays the user's summary, including their ARN, creation date, and console access status. The 'Permissions' tab is selected, showing a table of permissions policies. One policy, 'AdministratorAccess', is listed as being attached via the 'Admin' group. A 'Permissions boundary' section at the bottom indicates it is not set.

The screenshot shows the AWS IAM Dashboard. The left sidebar is the same as the previous screenshot. The main content area displays the 'IAM Dashboard' with a 'Security recommendations' section. This section shows three recommendations, all with green checkmarks indicating they are compliant: 'Root user has MFA', 'You have MFA', and 'Your user, SimeonLab-Admin, does not have any active access keys that have been unused for more than a year'.

Step 3: Set Up Cost Protection

A. Free Tier Usage Alert

- Go to **Billing** → **Preferences** → Check **Receive Free Tier Usage Alerts**

The screenshot shows the AWS Billing preferences page. The left sidebar contains navigation links for Billing preferences, Invoice delivery preferences, and Alert preferences. The main content area displays the 'Billing preferences' section. Under 'Alert preferences', there are two sections: 'AWS Free Tier alerts' and 'CloudWatch billing alerts'. Both sections show a green checkmark and the text 'Delivered to', indicating that alerts are enabled and delivered to the specified email address.

B. CloudWatch Billing Alarm (\$5 threshold)

- Switch to **us-east-1**
- Go to **CloudWatch** → **Alarms** → **Billing** → **Create alarm**
 - Metric: EstimatedCharges
 - Statistic: Maximum
 - Threshold: Static → Greater than → 5
 - Create new SNS topic → Add your email → Create alarm
- Confirm subscription email from AWS

The screenshot shows the AWS CloudWatch Alarms console. A green banner at the top states "Successfully created alarm \$5 Billing Alarm." with a "View alarm" button. Below the banner, the "Alarms" tab is selected, showing a list of alarms. The list contains one alarm, "\$5 Billing Alarm", which is in the "Insufficient data" state. The table columns include Name, State, Actions, Actions muted, Last state update (UTC), and Condition.

Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
\$5 Billing Alarm	Insufficient data	Actions enabled	-	2026-05-17 22:51:54	Estimated hours

C. AWS Budgets (Recommended)

- Go to **Budgets** → **Create budget**
 - Type: Cost budget
 - Name: Monthly-Learning-Budget
 - Budgeted amount: 10 USD
 - Alerts: 50%, 80%, 100%

The screenshot shows the AWS Budgets console. A green banner at the top states "Your budget Monthly-Learning-Budget has been created successfully." with a "Submit feedback" button. Below the banner, the "Budgets" tab is selected, showing a list of budgets. The list contains one budget, "Monthly-Learning-Budget", which is in the "OK" state. The table columns include Name, Thresholds, Health status, Billing View, Budget, Amount, and Forecast.

Name	Thresholds	Health status	Billing View	Budget	Amount	Forecast
Monthly-Learning-Budget	OK	Healthy	Primary View	\$10.00	-	-

Verification: You should now have:

- MFA on Root
- IAM Admin user
- At least one billing alarm

Phase 2: Infrastructure Awareness

Step 4: Launch a Free Tier EC2 Instance

1. Switch to **af-south-1** (Cape Town) — closest to Nigeria
2. Go to **EC2** → **Launch instance**
 - Name: Week1-Capstone-WebServer
 - AMI: **Amazon Linux 2023**
 - Instance type: **t3.micro**
 - Key pair: Create new (capstone-key)
 - Network settings:
 - Create new Security Group
 - Allow SSH (port 22) from **My IP** (for security)
 - Allow HTTP (port 80) from Anywhere (0.0.0.0/0) — for web server
 - Storage: Default 8GB gp3
 - **Launch**

Instance summary for i-08334fff80883bed2 (Week1-Capstone-WebServer) [Info](#)

Updated less than a minute ago

Instance ID  i-08334fff80883bed2	Public IPv4 address  13.247.55.226 open address 
IPv6 address —	Instance state  Running
Hostname type IP name: ip-172-31-12-89.af-south-1.compute.internal	Private IP DNS name (IPv4 only)  ip-172-31-12-89.af-south-1.compute.internal
Answer private resource DNS name IPv4 (A)	Instance type t3.micro
Auto-assigned IP address  13.247.55.226 [Public IP]	VPC ID  vpc-07b1ea53eda8ada8d 
IAM role —	Subnet ID  subnet-0655ea282f8807b6e 

3. Wait until **Status check = 2/2 checks passed**

Instances (1) Info							
Saved filter sets Choose filter set 		Find Instance by attribute or tag (case-sensitive)		   Instance state  Actions  Launch instances 			
 Name 		 Instance ID		 Instance state		 Instance type	
 Week1-Capsto...		i-0f0212d234089d027		 Running 		t3.micro	
				 3/3 checks passed		af-south-1a	
						ec2-16-28-27-67.af-sou...	
						1	

Phase 3: Connect & Configure (Hands-on Cloud)

Step 5: Connect to your instance

- Download the .pem key
- On your local terminal (Git Bash / WSL / Mac Terminal):

Bash

```
chmod 400 capstone-key.pem
```

```
ssh -i "capstone-key.pem" ec2-user@<your-public-ip>
```

```
ec2-user@ip-172-31-12-89 ~$
```

```
igesl@ZSER MINGW64 ~/OneDrive/Desktop/New folder
$ chmod 400 "capstone-key.pem"
```

```
igesl@ZSER MINGW64 ~/OneDrive/Desktop/New folder
$ ssh -i "capstone-key.pem" ec2-user@ec2-13-247-55-226.af-south-1.compute.amazonaws.com
```

```
The authenticity of host 'ec2-13-247-55-226.af-south-1.compute.amazonaws.com (13.247.55.226)' can't be established.
```

```
ED25519 key fingerprint is: SHA256:9aUWO9tGKaQ868y8wNLHNWljAdnq4ESBL/aNm/Rsqxo
```

```
This key is not known by any other names.
```

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Warning: Permanently added 'ec2-13-247-55-226.af-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
```

```
** WARNING: connection is not using a post-quantum key exchange algorithm.
```

```
** This session may be vulnerable to "store now, decrypt later" attacks.
```

```
** The server may need to be upgraded. See https://openssh.com/pq.html
```

```
_#_
~\##~ Amazon Linux 2023
~~\#####
~~\####|
~~\###|
~~\#/
~~V~'|'-> https://aws.amazon.com/linux/amazon-linux-2023
~~~~
~~~.-.-
~~~/ / \
[ec2-user@ip-172-31-12-89 ~]$
```

Step 6: Deploy a simple web server (Apache)

Bash

```
sudo yum update -y
```

```
sudo yum install httpd -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

Create a simple Nigerian-themed page

```
sudo bash -c 'cat > /var/www/html/index.html << EOF
```

```
<!DOCTYPE html>
```

<html>

```
<head><title>Welcome from Lagos</title></head>
```

```
<body style="text-align:center; margin-top:50px; font-family:Arial;">
```

Week 1 Capstone Complete!

<h2>I am running in af-south-1 (Cape Town)</h2>

<p>Built by Your Name • BaseStack Academy</p>

</body>

</html>

EOF'

sudo systemctl restart httpd

```
ec2-user@ip-172-31-12-89:~$ sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd

# Create a simple Nigerian-themed page
sudo bash -c 'cat > /var/www/html/index.html << EOF
<!DOCTYPE html>
<html>
<head><title>Welcome from Lagos</title></head>
<body style="text-align:center; margin-top:50px; font-family:Arial;">
  <h1>🇳🇬 Week 1 Capstone Complete!</h1>
  <h2>I am running in af-south-1 (Cape Town)</h2>
  <p>Built by <strong>Simeon Siaka</strong> • BaseStack Academy</p>
</body>
</html>
EOF'

sudo systemctl restart httpd
Amazon Linux 2023 Kernel Livepatch repository                               312 kB/s | 38 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Sun May 17 23:25:40 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
---------	--------------	---------	------------	------

```
ec2-user@ip-172-31-12-89:~$ sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd

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Amazon Linux 2023 Kernel Livepatch repository                               312 kB/s | 38 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Sun May 17 23:25:40 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	47 k
Installing dependencies:	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
apr-util-ldap	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
generic-logos-httpd	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-core	noarch	2.4.66-1.amzn2023.0.1	amazonlinux	13 k
httpd-filesystem	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	81 k
httpd-tools	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k
libbrotli	x86_64	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	15 k
apr-util-openssl	x86_64	2.0.27-1.amzn2023.0.3	amazonlinux	166 k
mod_http2	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	60 k
mod_lua	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	60 k

```
Transaction Summary
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                          2.6 MB/s | 97 kB    00:00
(2/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm                    322 kB/s | 13 kB    00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                              2.7 MB/s | 129 kB    00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm                 562 kB/s | 15 kB    00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm            731 kB/s | 19 kB    00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                          1.8 MB/s | 47 kB    00:00
(7/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm                     3.3 MB/s | 81 kB    00:00
(8/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch.rpm                414 kB/s | 13 kB    00:00
(9/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm                      29 MB/s | 1.4 MB    00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm                       11 MB/s | 315 kB    00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                        1.1 MB/s | 33 kB    00:00
```

```
ec2-user@ip-172-31-12-89:~$ sudo yum install httpd
Installing      : apr-util-1.6.3-1.amzn2023.0.2.x86_64                4/13
Installing      : mailcap-2.1.49-3.amzn2023.0.3.noarch                5/13
Installing      : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64           6/13
Installing      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64              7/13
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch      8/13
Installing      : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch      8/13
Installing      : httpd-core-2.4.66-1.amzn2023.0.1.x86_64           9/13
Installing      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64            10/13
Installing      : mod_lua-2.4.66-1.amzn2023.0.1.x86_64              11/13
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  12/13
Installing      : httpd-2.4.66-1.amzn2023.0.1.x86_64                13/13
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64                13/13
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64                   1/13
Verifying       : apr-util-1.6.3-1.amzn2023.0.2.x86_64              2/13
Verifying       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64         3/13
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64     4/13
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  5/13
Verifying       : httpd-2.4.66-1.amzn2023.0.1.x86_64               6/13
Verifying       : httpd-core-2.4.66-1.amzn2023.0.1.x86_64          7/13
Verifying       : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch     8/13
Verifying       : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64         9/13
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64            10/13
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch              11/13
Verifying       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64           12/13
Verifying       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64             13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64
  apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mod_http2-2.0.27-1.amzn2023.0.3.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-12-89 ~]$
```

Open your browser: <http://<your-public-ip>>

You should see your page!



 **Week 1 Capstone Complete!**

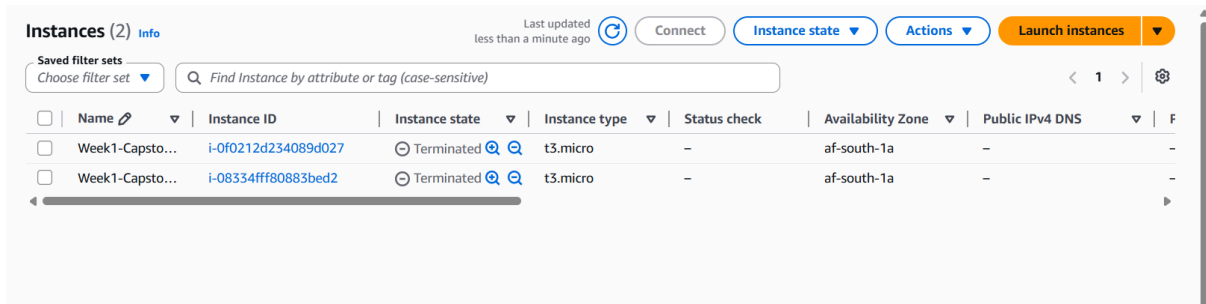
I am running in af-south-1 (Cape Town)

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Phase 4: Cleanup + Economics Check

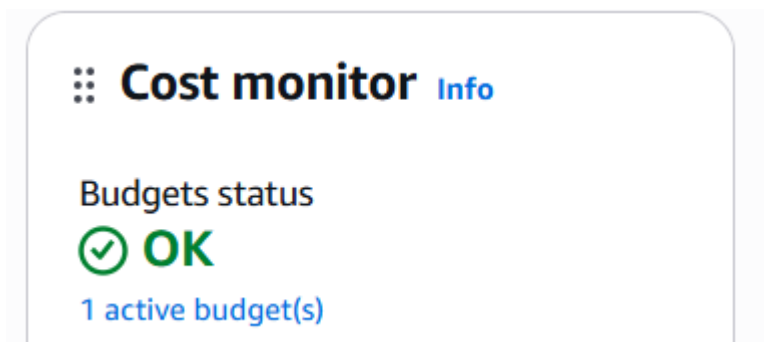
Step 7: Stop/Terminate Instance (Very Important)

- Go back to EC2 console
- Select instance → **Instance state** → **Stop** (you can start later)
- Or **Terminate** if you're done



Step 8: Final Cost Check

- Go to **Billing** → **Cost Explorer** (or **Budgets**)
- Confirm your spend is still near zero



Thank you